SR-SEISMIC: RESCUE SOFTWARE AND AUTOMATED WIRELESS BEACON FOR SMARTWATCHES

Technical Architecture Memory & Copyright Declaration - GitHub Document v3.0

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PART I: DEVELOPMENT FLOWCHART (THE WHAT)

The core software operates under a strict chronological protocol designed for structural collapse scenarios caused by earthquakes, sequencing power management and telemetry from initial impact to victim extraction.

1. **Earthquake Alert Reception:** Real-time official notification acquisition (e.g., *Google Earthquake Alerts System*), triggering high-priority background initialization.
2. **Black Box Mode (Impact Telemetry):** Autonomous local storage (SQLite/Room) of continuous data from accelerometers, gyroscopes, barometers, and pulse sensors to log structural trauma.
3. **Deep Sleep Transition:** Following an explicit expert-defined safety window to allow cellular/Wi-Fi use by conscious survivors, non-critical services are halted to preserve capacity.
4. **Beacon System Initialization:** Omnidirectional Bluetooth Low Energy (BLE) advertisement broadcasting containing User ID, biometric constants, and barometric telemetry.
5. **Intermittent Duty Cycle:** Periodic cycling between deep sleep and transmission bursts, balanced to provide a 72-to-96-hour runtime.
6. **Rescuer Capture:** Active parsing of open BLE beacons via the rescuer app without physical pairing, compiling structural victim databases.
7. **Automated Triage:** Immediate priority classification (Fatalities, Stable Trapped, Critical Trapped) using raw heart rate and oximetry variance.
8. **"I See You" Handshake:** Target-directed acknowledgment packet (ACK_TE_VI) modifying the device internal clock to increase beacon frequency (e.g., every 15s) for micro-localization.
9. **Proximity Telemetry Tracking:** Received Signal Strength Indicator (RSSI) tracking to map horizontal X and Y surface axes.
10. **Active Radar (Z-Depth & Density Calibration):** High-precision depth calculation using Time of Flight ($ToF = T_{total} - T_{proc}$) and power attenuation ($\Delta P = P_{emitted} - P_{received}$) to distinguish void air gaps from solid concrete layers.

11. **Psychological Comfort Dispatch:** Forced micro-speaker execution rendering high-priority localized voice output to reassure the victim.

PART II: TECHNICAL FOUNDATIONS & FEASIBILITY ANALYSIS (THE HOW)

1. Hardware Overclocking & Thermal Management

By dropping the duty cycle into micro-bursts, transient thermodynamic boundaries change. AMOLED overdriven brightness ($T_{on}=50\text{ ms}$) delivers massive lumen projection without damaging substrates, satisfying $E_{th} = I^2 \times R \times T_{on}$ due to the massive cooling delay ($T_{off}=600.000\text{ ms}$). Acoustic coils utilize pure transient sine/square burst patterns to reach maximum mechanical displacement without Joule heating failure.

2. Acoustic Propagation Optimization

The audio signature is locked to **3.5 kHz - 4 kHz** for Triple Coincidences: isolating signals from heavy machinery noise (<1 kHz) via digital Band-pass Filters, matching canine (K9) high-frequency neural thresholds, and capitalizing on the natural 2.5 cm human ear canal resonance to gain passive **10 to 15 dB** amplification without drawing current.